Structures 57 (2023) 105336

Contents lists available at ScienceDirect

Structures

journal homepage: www.elsevier.com/locate/structures

Experimental studied on suppressing vortex-induced vibration of closed-box girder bridges with various moveable guide vanes

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ARTICLE INFO

Keywords:
Closed-box girder
Moveable guide vane
Vortex-induced vibration control
Movement characteristics

ABSTRACT

A new moveable guide vane with the combination of the spring and guide vane was proposed to suppress the vortex-induced vibration (VIV) of closed-box girder bridges. This paper studies the influence of two important parameters (*i.e.,* horizontal length and spring stiffness) of a moveable guide vane on controlling the VIV of a super long-span suspension bridge with closed-box girder by using VIV control tests. The VIV test of 1:50 scaled ratio of sectional model of the original closed-box girder under three wind attack angles were firstly conducted. The suppression ratios of vertical and torsional VIV displacements of the closed-box girder as well as the movement characteristics of moveable guide vane were then analyzed, respectively. The results show that the movable guide vane could significantly reduced the Root Mean Square (RMS) values of VIV displacement of the closed-box girder are after installing, especial for the torsional VIV displacement. A moveable guide vane with a relative smaller horizontal lengths or a relative larger spring stiffness could produce a better outcome of VIV control. The double VIV amplitude of the moveable guide vanes results in the energy dissipation of the closed-box girder in the lock-in VIV wind speed region, while the vertical displcement of the moveable guide vane are close to those of the closed-box girder in non-VIV wind speed region. Besides, the nonlinear exponential fitting function could effectively reflect the relationship among the wind speed, the VIV displacement of the MGV and the closed-box girder.

1. Introduction

As a common main grider of suspension bridges, the streamlined closed-box girder has the advantage of large section rigidity and good overall stability, as well as has great aerodynamic performance [1,2]. Since accessory components (*e.g.* deck railings) could passivate the aerodynamic shape of the closed-box girder, the vortex-induced vibration (VIV) phenomenon of closed-box girder bridges were observed all around the world [3]. For example, the VIV event of Storebælt suspension bridge with the maximum vertical displacement of about 0.32 m [4], the well-known VIV event of Humen suspension bridge with the maximum vertical acceleration of 1.3 m/s2 and displacement of about 0.31 m [5]. Therefore, It is necessary to adopt effectivley VIV control measures to eliminate the occurrence of VIV in long-span suspension bridge with closed-box girders.

Compared with the active aerodynamic countermeasure, passive aerodynamic countermeasure has the advantages of excellent economic and simple structure, which are widely used to suppress VIV in many practical bridge projects in the last decade [6,7]. Through wind tunnel testing and CFD simulation, the control effects of different aerodynamic countermeasures (*i.e.*, the central slot [8]), the vertical stablizer [9], the spoilers on crash barriers [10], the wavy railings [11]), the ventilation rates of different railings [12], the airflow-depressing board [12], the grid plates [13], the different web modifications[14] on the VIV performance of bridges are investigated, respectively. Moreover, the guide vanes of 8 mm plate struts were effectuated along the bottom side panel joint of the Storebælt suspension bridge girder cross section at 2 m intervals [4] and the guide vane was installed below the joints of the horizontal and the inclines bottom plates of Gjemnessund bridge [15]. Although the immoveable guide vane is one of effective VIV countermeasures in the airflow separation around the box girder, their control effectiveness mainly depends on their aerodynamic configurations and location [16–18].

In recent years, moveable aerodynamic countermeasures are

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https://doi.org/10.1016/j.istruc.2023.105336
Received 12 July 2023; Received in revised form 28 September 2023; Accepted 30 September 2023
Available online 8 October 2023

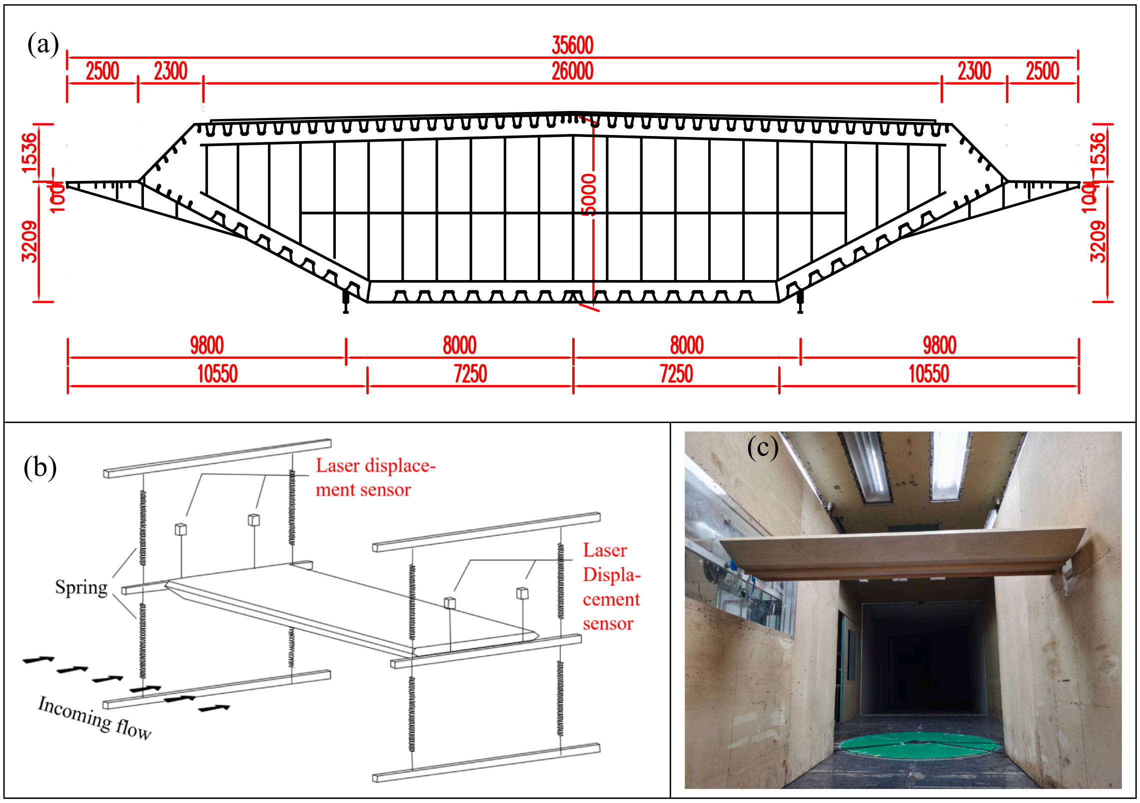

Fig. 1. The VIV tests of the closed-box girder without the MGV: (a) the cross section dimension; (b-c) experimental setup of the closed-box girder.

Table 1
Geometrical and mechanical parameters of the real bridge and sectional model.

Properties		Similarity ratio	Real bridge	Section model
Geometric scale	Length L/m	$\lambda_L = 1{:}50$	87	1.74
	Width B/m	$\lambda_L = 1{:}50$	35.6	0.712
	Height H/m	$\lambda_L = 1{:}50$	5.0	0.1
Equivalent mass	Mass/unit length m /kg/m	$\lambda_m = 1{:}50^2$	30,645.4	12.258
	Mass moment of inertia/unit length I_m/ kg·m^2/m	$\lambda_{jm} = 1{:}50^4$	4,738,390	0.758
Frequency	Vertical f_v /Hz	$\lambda_\omega = 25{:}1$	0.1132	2.830
	Torsional f_t/Hz	$\lambda_\omega = 25{:}1$	0.2757	6.893
Wind speed	Standard Design Wind Speed	$\lambda_w = 1{:}2$	58.7	29.35

proposed to improve the aerodynamic control effects of bridges, such as, a moveable vertical stabilzer with the combiantion of a tuned mass damper (TMD) and two vertical quartile plates [19], a mixed passive control system of combining the leading and trailing edge flaps and TMD [20], a self-issuing jet [21], eccentric-wing [22], inerter-based dampers [23–24]and the TMD with a mass polar moment of inertia's optimum configuration [25] to increase the critical flutter wind speed by altering the incoming airflow. Furthermore, the movable guide vane (MGV) with the TMD and guide vane designed to improve the VIV control effect of twin-box girder bridges, in which the MGV with large shorter horizontal length and small spring stiffness has more control effect of TMD than that of the immovable guide vane, and the coulping effect of the TMD and immovable guide vane could significantly suppress the VIV of the twin-box girder by the about double amplitude in vertical movement of the MGV itself [26]. However, the influence of the MGV on VIV performance of closed-box girders has not been fully understood so far.

In order to further application in suspension bridges of movable guide vanes, the present study the structural parametric optimization of the movable guide vane on VIV performance of closed-box girders by a series of wind tunnel tests. At first, the VIV tests were conducted on the 1:50 sectional models of the original closed-box girder and their VIV responses are analyzed. Secondly, the movable guide vanes were designed and the control effects of two critical parameters of the moveable guide vanes (*i.e.,*three horizontal length and three spring stiffness) were then optimizated. Finally, the movement characteristics of the closed-box girder and movable guide vane are also investigated.

2. Viv tests of the original Closed-Box girders

2.1. VIV tests of the orginal closed-box girder without the MGV

As shown in Fig. 1(a), a typical closed-box girder suspension bridge with mian span of 1756 m was selected to study the VIV control effect, in which the width and heigth of the closed-box girder is 35.6 m and 5.0 m. Fig. 1(b-c) shows the experimental setup of the closed-box girder without the movable guide vanes, which were conducted in the TJ-2 Boundary Layer Wind Tunnel, Tongji University. There are four laser displacement transducers were used for measuring the vertical and torsional displacement of the main girder in Fig. 1(b-c). Table 1 shows the structural parameters of the real bridge and sectional model with a geometric scale ratio of 1:50. The first torsional and total mass of the sectional model is 6.893 Hz and 21.33 kg, respectively.

2.2. VIV responses of the original closed-box girder

According to the Wind-Resistance Design Specification for Highway Bridges of China (JTG/T D60-01–2018), the amplitude limitations of the vertical and torsional VIV of the real bridge are defined in Eq.(1–2) as follows:

$$0.04/ = 0.04/0.1132 = 0.35m \tag{1}$$

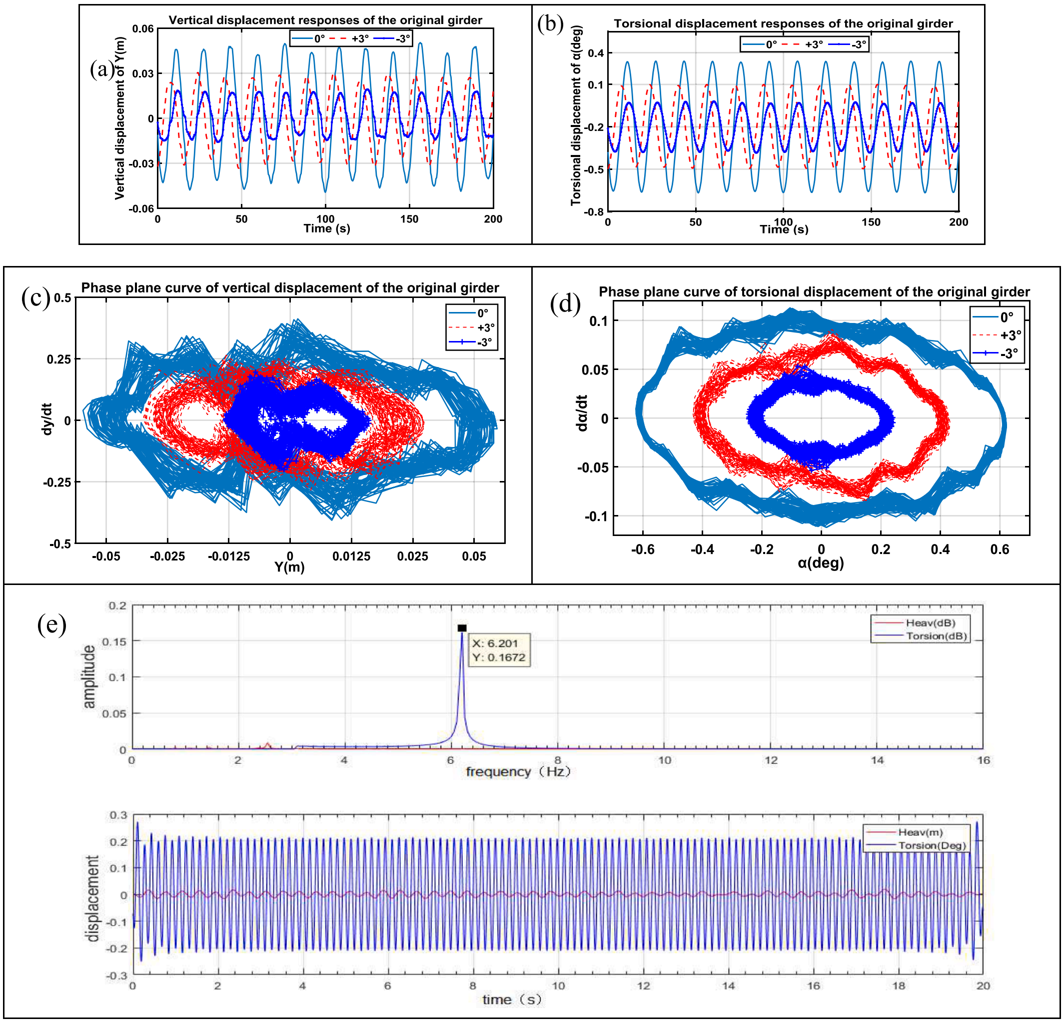

Fig. 2. VIV responses of the original closed-box gider: (a-b) vertical and torsional displacement responses; (c-d) phase plane curves of vertical and torsional displacement responses; (e) the time-history and spectrum of VIV displacement at the wind speed of 4.8 m/s.

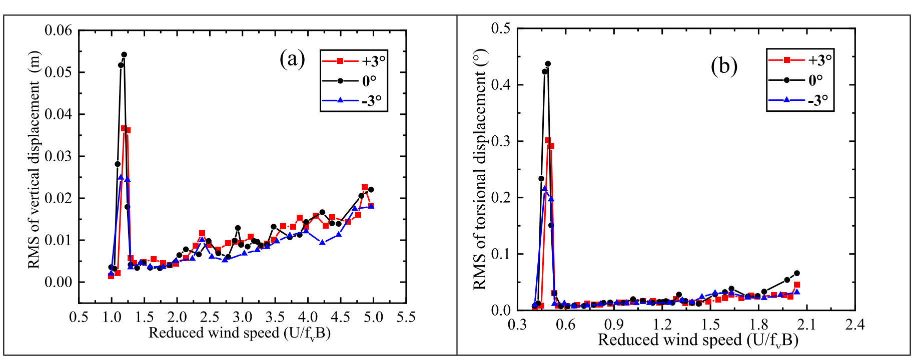

Fig. 3. VIV responses of the original closed-box gider:(a-b) Vertical and Torsional displacement.

Table 2
The s_t number of VIV in lock-in wind speed region.

Wind speed (m/s)	4.2	4.4	4.8	5.2	5.6
S_t	0.134	0.128	0.117	0.108	0.101

$$4.56 = 4.56/(35.6 \times 0.2757) = 0.46^{\circ} \tag{2}$$

where B is the width of the closed-box girder, the f_v and f_t are the vertical and torsional frequency of the bridge, respectively.

The vertical and torsional displacement responses and their phase plates of these three wind attack angles at the lock-in wind speeds of 4.4 m/s and 5.0 m/s are displayed in Fig. 2(a-d), respectively. The irregular sinusoidal manner of vertical displacement responses produce the irregular circular shape of phase planes, while the limit cycle oscillation

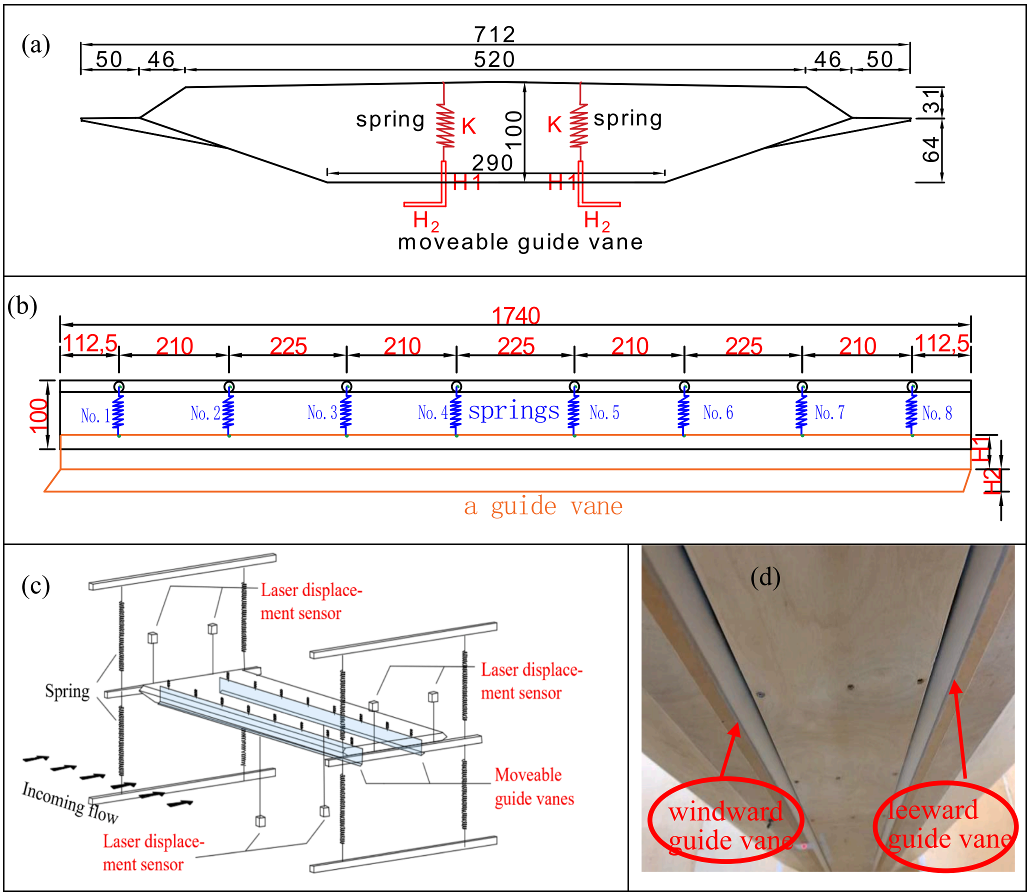

Fig. 4. The VIV tests of the closed-box girder. (a) dimension of MGV; (b) arrangement of MGV; (c-d) experiment setup of MGV (unit: mm).

Table 3
VIV control testing cases of various MGV.

Case	Number × dimension of spring	Total spring stiffness K (g/mm)	Vertical height H_1 and horizontal height H_2 (mm)	Wind attack angle (°)
1	–	–	–	0,±3
2	3 × 0.7	24	60 + 10	0,±3
3	3 × 0.7	24	60 + 20	0,±3
4	3 × 0.7	24	60 + 30	0,±3
5	2 × 0.7 + 2 × 0.5	18	60 + 10	0,±3
6	5 × 0.6	29	60 + 10	0,±3

(LCO) in a sinusoidal manner was observed in torsional displacement responses under three wind attack angles. All the torsional phase planes regularly develop from inside toward outside in elliptical shapes, the elliptical area under the α = 0° is the largest and the elliptical area under α=+3° is the smallest. In particular, the time history and spectrum of the maximum vertical and torsional displacement responses under the wind speed of 4.8 m/s are presented in Fig. 2(e), in which the vertical and torsional stable sinusoidal motion have the peak frequencies of about 2.5 Hz and 6.2 Hz, respectively.

Fig. 3 illustrates the RMS values of vertical and torsiaonal displacement responses of the original closed-box girder bridge, respectively. Under three typical wind attack angles (α = -3°, 0° and + 3°), the α = 0° and α = -3° is the most unfavorable and favorable wind attack angle, respectivley. As for the vertcial VIV responses, their maximum RMS values is about 0.055 m at the reduced wind speed of U/f_vB from 1 to 1.3, while the maximum torsional RMS values of are about 0.46° from U/f_vB = 0.4 to 0.6. Table 2 also shows that the Strouhal number (St) decreases from 0.134 to 0.101 when the lock-in wind speed increases from 4.2 m/s to 5.6 m/s. Thus, it is necessary to adopt the effective moveable guide vane to suppress the torsional VIV of the bridge.

3. Viv control effects of various movable guide vanes

3.1. VIV control tests with vaious movable guide vanes

The MGV is composited of the immovable guide vane and the spring, in which many springs are connected by the top plate of the main girder and the upper end of guide vane. Based on the Den Hartog optimization theory in Eq. (3) [27–29], the spring stiffness and length of the MGV are determined by the Eq.(4):

$$\mu = \frac{m_2}{m_1}, \gamma = \frac{1}{1+\mu}, \omega_2 = \frac{\mu}{1+\mu}\omega_1 \tag{3}$$

$$k_2 = m_2{\omega_2}^2, H_1 + H_2 = \frac{m_2}{\rho_2 LD} \tag{4}$$

where, m_1 and ω_1 are the mass and frequency of the section model of the closed-box girder, respectively. γ is the most optimal frequency ratio. m_2, k_2, and ω_2 are the mass, stiffness, and frequency of the guide vane, respectively. ρ_2 is the plexiglass density of the guide vane. The H_1 and H_2 are the vertical and horzontial height of the guide vane. The L, and D is the total length and depth of the guide vane is designed to 1.74 m and 0.005 m, respectively.

Since the first torsional frequency has great contribution on the torsional vortex-induced vibration, we aim to suppress the first torsional frequency to calculate the optimal mass and stiffness of the MGV. According to the mass and frequency of the MGV in the Eq.(3–4), the length and spring stiffness of the MGV are 0.01 m and 240 N/m, respectively. In

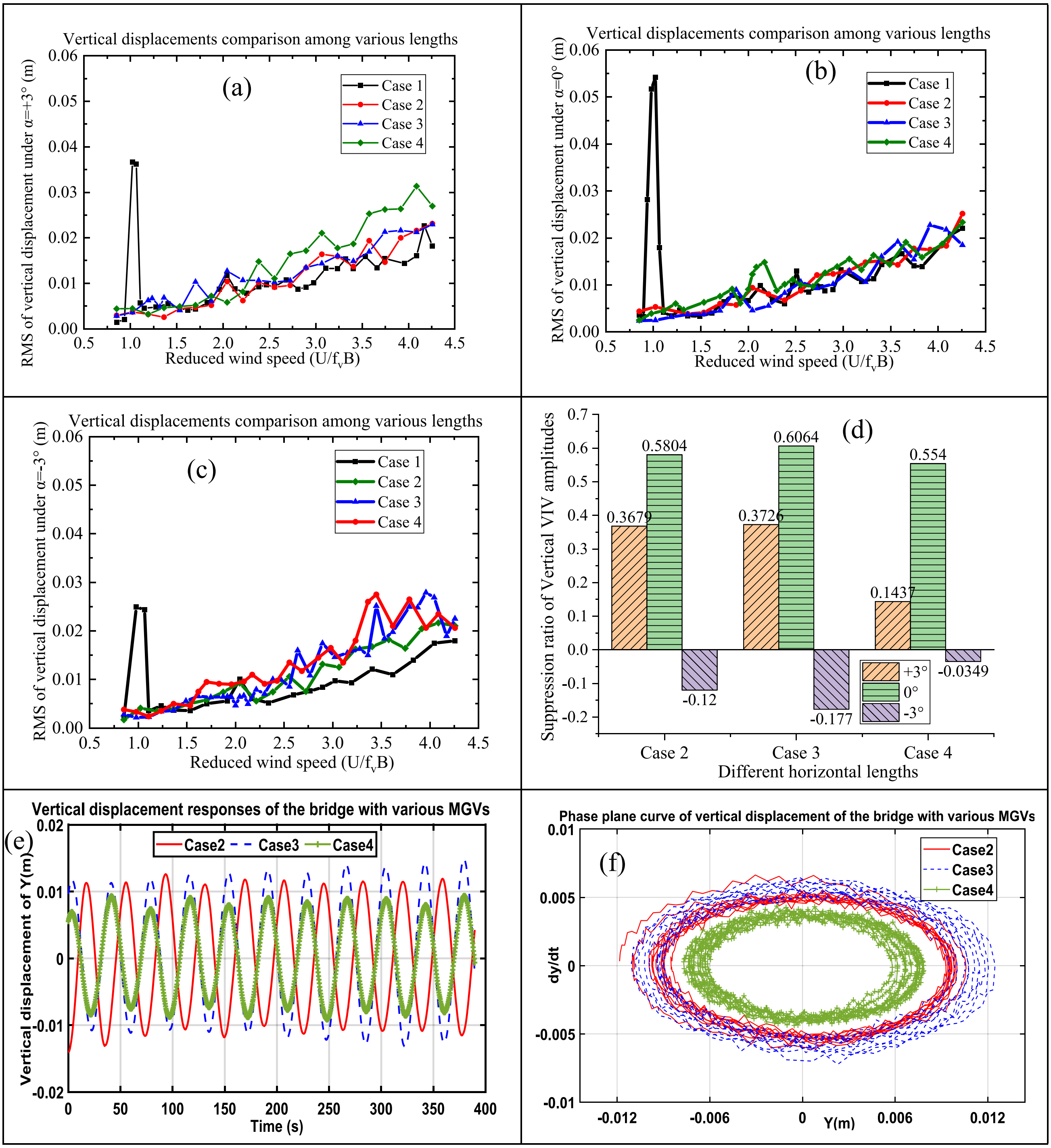

Fig. 5. Vertical VIV responses of the bridge with various horizontal lengths of MGV: (a) α=+3°,(b) α=+0°,(c) α = -3°, (d) suppression ratios of three wind attack angles, and (e-f) vertical VIV displacement responses and their phase plane curves under α = -3°.

order to conveniently replace the spring with different stiffnesses inside the closed-box girder, the MGV is designed to the quarter point position of bottom plate instead of the corner, and these guide vanes still effectively change the vortices distribution and motion in the downstream of the main girder to alter the frequency of vortex shedding. As shown in Fig. 4, there are eight springs connected with the top plate of the sectional model and the upper of a guide vane with the interval of about 0.21 m along the length of section model. What is important, two laser displacement transducers are also used to measure the vertical displacement of the windward and leeward guide vane, respectively.

In order to investigate the influence of the spring stiffness (K) and horizontal length (H_2) of the MGV on the VIV performance of the closed-box girder bridge, three representative horizontal lengths (L = 10 mm, 20 mm and 30 mm) and three representative spring stiffnesses (K = 180 N/m, 240 N/m, and 290 N/m) were designed. As listed in Table 3, there are six testing cases of the MGV, including three different horizontal lengths (Case 2 to 4), and three different spring stiffnesses (Case 2,5 and 6).

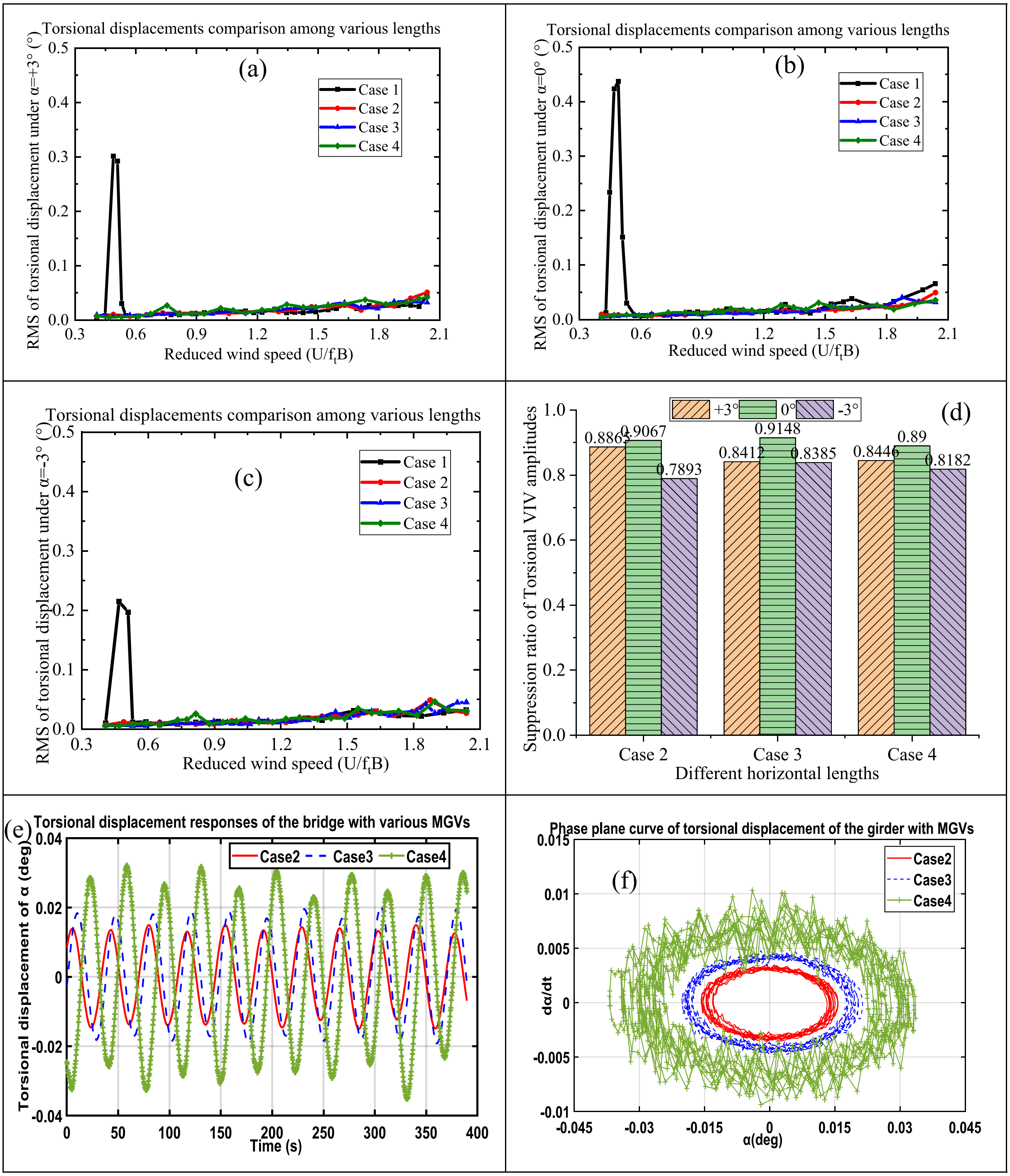

Fig. 6. Torsional VIV responses of the bridge with various horizontal lengths of MGV: (a) α=+3°,(b) α=+0°,(c) α = -3°, (d) suppression ratios of three wind attack angles, and (e-f) torsional VIV displacement responses and their phase curves under α=+3°.

3.2. VIV responses under different horizontal heights of moveable guide vanes

In order to compare the influence of the horizontal heights of the MGV on the VIV control effect, vertical VIV displacement responses and their phase plane curves of the closed-box giders bridge with three different horizontal heights are illustrated in Fig. 5, respectively. Fig. 5 describes that the MGV effectively suppress the vertical VIV responses in the lock-in wind speed region from 4 m/s to 6 m/s. Although the vertical displacement responses gradually become larger with the increase of wind speed, ther are not obvious high-order VIV phenomenon. Under the wind attack angles of α = 0° and + 3°, the RMS of vertical displacement overall increases with the increase of the horizontal height, since the longer horizontal lengths of the moveable guide vane leads to a more bluff body of the closed-box girder. Fig. 5(d) compares the suppression ratio (φ) of vertical VIV amplitude without MGV (Case 1) and with moveable guide vane (Case 2–4), respectively. $\varphi = \frac{A_1 - A_o}{A_1} \times 100\%$, where the A_1 and A_o is the maximum values of VIV displacement of the Case 1 and other cases, respectively. The suppression ratios of Case 3 are slightly larger than those of the Case 2 and Case3 under the α = 0° and α=+3°, the suppression ratios of Case 3 are around 60.64 % and −17.7 % under the α = 0° and α = -3°, respectively. Fig. 5(e-f) shows

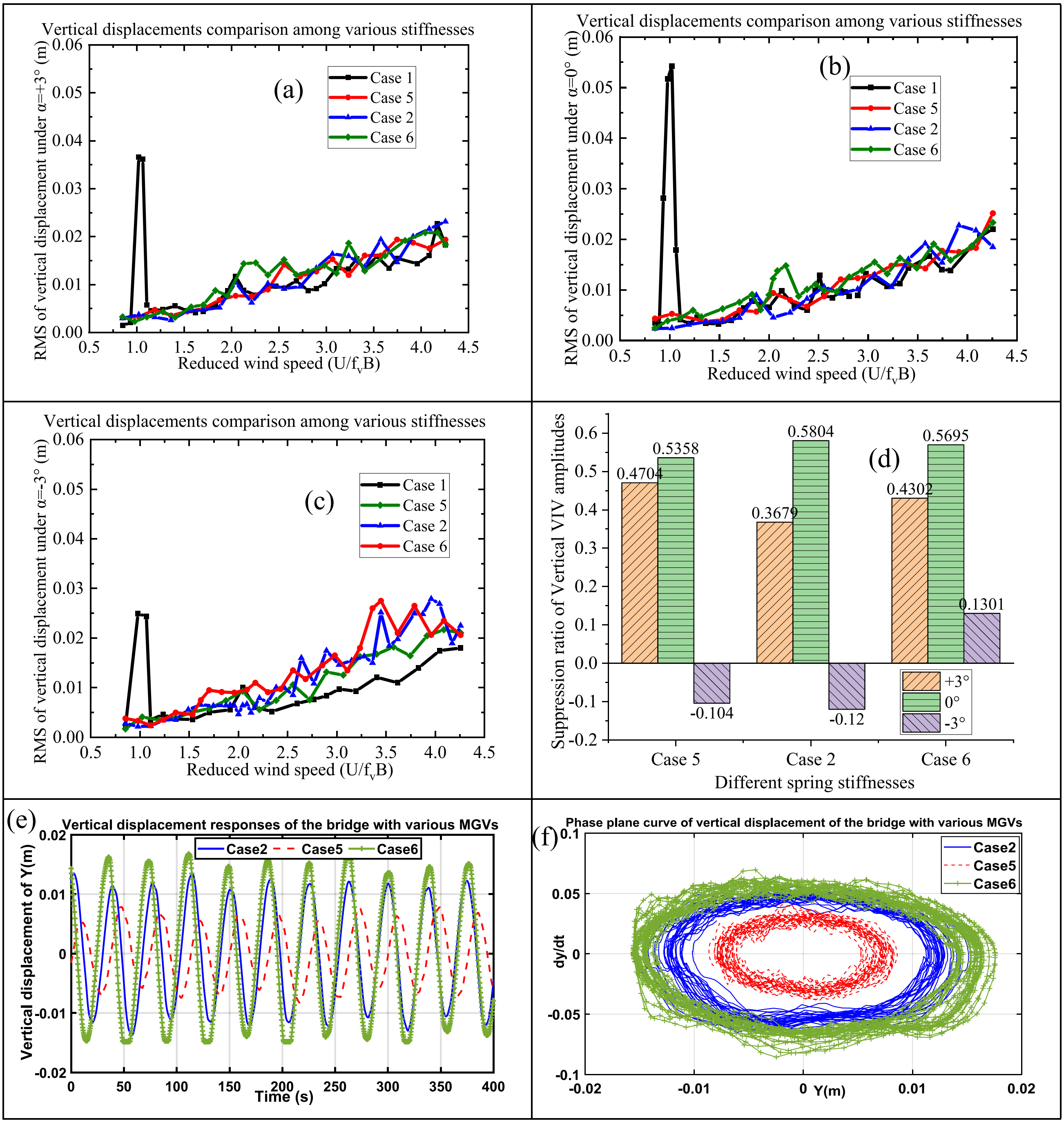

Fig. 7. Vertical VIV responses of the MGV with various spring stiffnesses: (a) α=+3°, (b) α=+0°, (c) α = -3°, (d) suppression ratios of three wind attack angles, and (e-f) vertical VIV displacement responses and their phase curves under α = 0°.

that the peak vertical amplitudes of the irregular sinusoidal motions for Case 2, Case 3, and Case4 are about 0.013,0.015 and 0.01, respectively. The cicular shapes' area of vertical phase planes for the Case 4 is much smaller than those of the Case 2 and Case 3.

The movable guide vane has obvious vertical dispalcement and their small horizontal displacement could be neglected when subjected to the relative small lock-in wind speed, and the horizontal motion movement of the MGV could be restricted by multiple springs and the bottom plate on both sides. Fig. 6 shows that the MGV has great effect on the torsional VIV suppression of the closed-box girder under three wind attack angles. There are a small torsional VIV responses of about 0.05° and 0.04° for the Case 4 under the α=+3° and α = -3°, respectively. In addition, Fig. 6 (d) indicates that the maximum suppression ratio of torsional displacement is the Case 3 with the 91.48 % and 83.85 % under the α = 0° and α = -3°, respectively. The peak torsional amplitudes of the irregular sinusoidal motions for Case 2, Case 3, and Case4 are about 0.016°,0.02° and 0.035° in Fig. 6(e-f), respectively. The cicular shapes' areas of torsional phase planes for the Case 4 and Case 2 are the largest and the smallest, respectively. Therefore, the installation of moveable guide vane can effectively decrease the VIV responses, especial for the torsional VIV responses, whereas the longer horizontal heights of the MGV could produce smaller VIV responses.

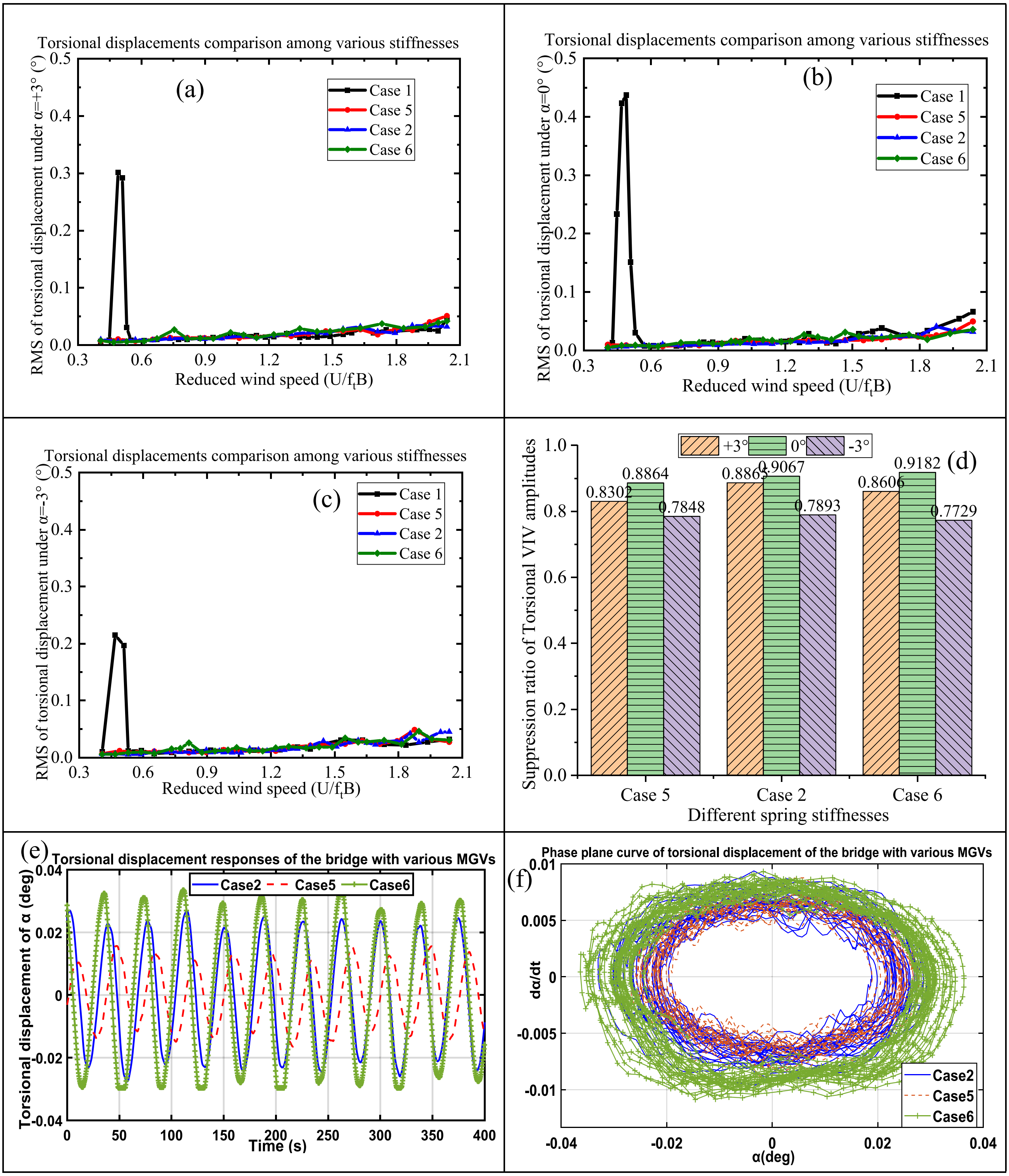

Fig. 8. Torsional VIV responses of the MGV with various spring stiffnesses: (a) α=+3°, (b) α=+0°,(c) α = -3°, (d) suppression ratios of three wind attack angles, and (e-f) vertical VIV displacement responses and their phase curves under α = 0°.

3.3. VIV responses under various spring stiffnesses of moveable guide vanes

The control effects of the spring stiffnesses of the MGV on vertical VIV displacement responses of the closed-box girder bridge are comapred in Fig. 7. Although the vertical displacement responses gradually become larger with the increase of wind speed, the MGV effectively suppress the vertical VIV responses in the lock-in wind speed region from 4 m/s to 6 m/s. The MGV with smaller spring stiffness (*e.g.*, Case 5) has the largest suppress ratio of about 47 % under the wind attack angles of α=+3°, while the MGV with larger spring stiffness (*e.g.*, Case 6) also has the positive suppress ratio of about 13 % under the wind attack angles of α = -3°. Fig. 7(e-f) shows the peak vertical amplitudes of the irregular sinusoidal motions for Case 2, Case 5, and Case 6 are about

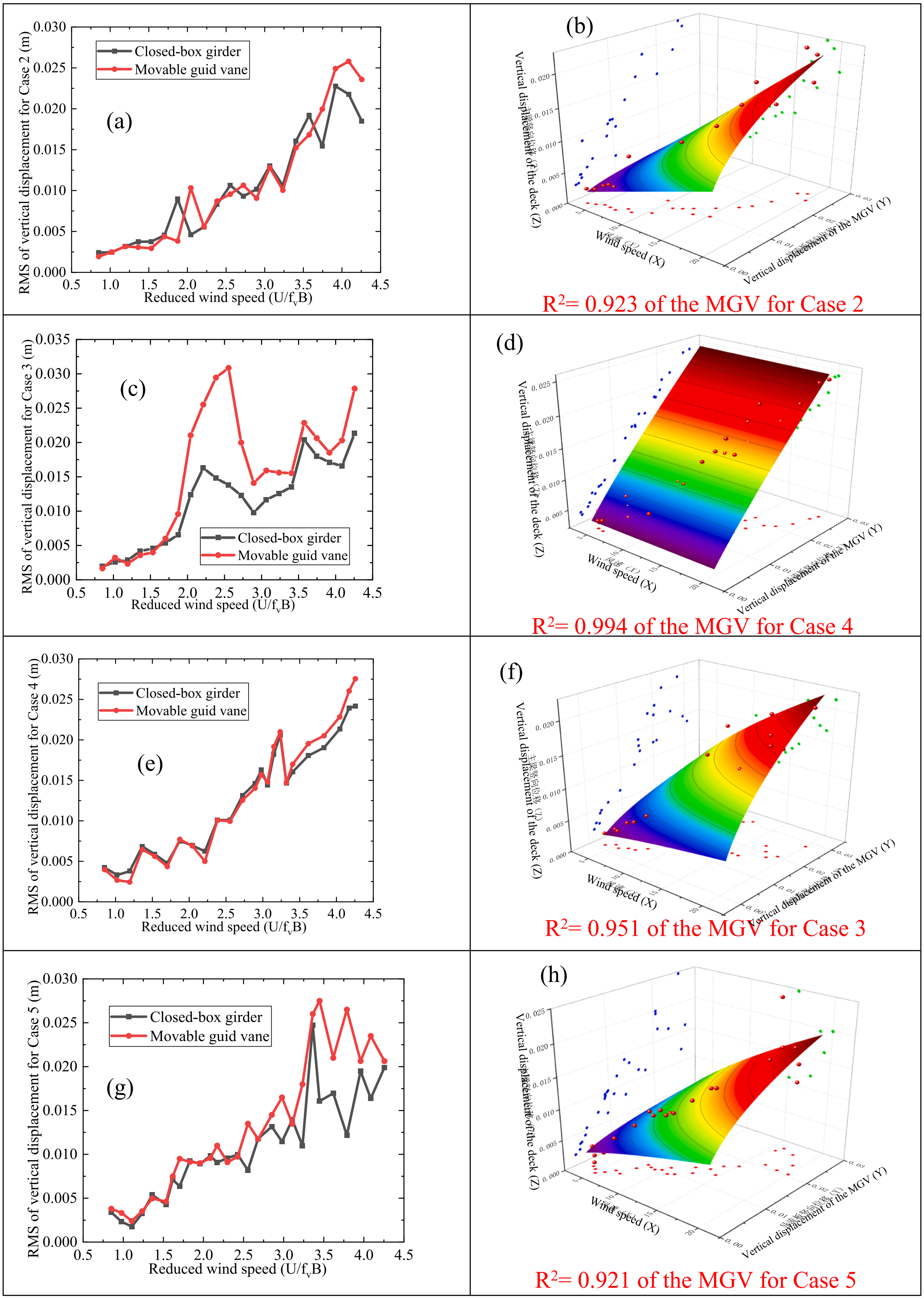

Fig. 9. Movement characteristics of vertical displacement of the closed-box girder and the mgv under α = 0°: (a-b) Case 2; (c-d) Case 3; (e-f) Case 4; (g-h) Case 5 (i-j) Case 6.

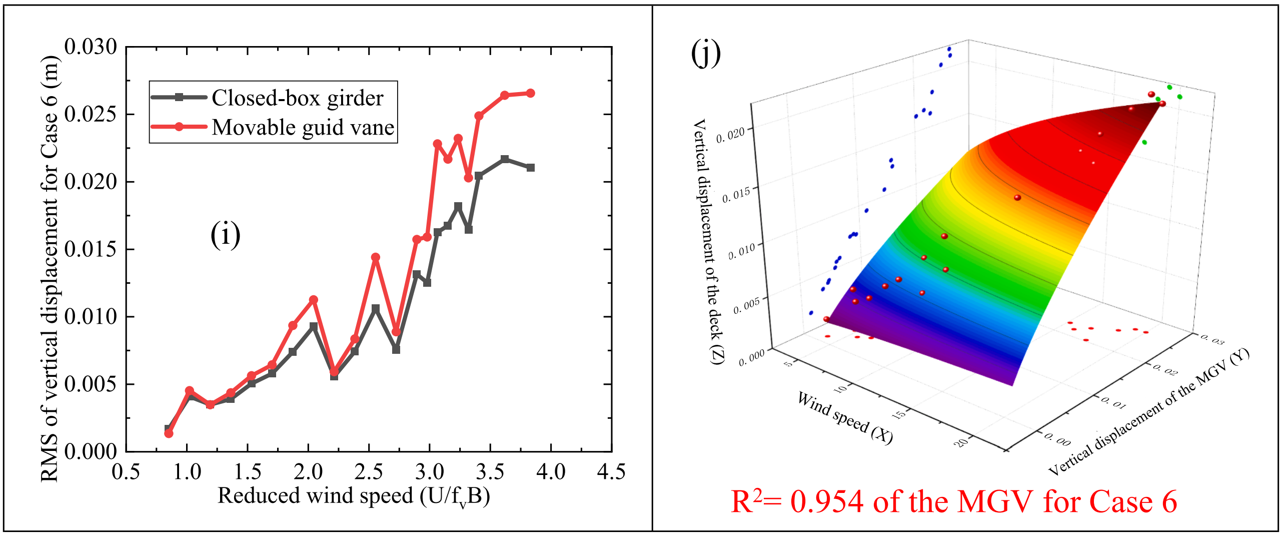

Fig. 9. (*continued*).

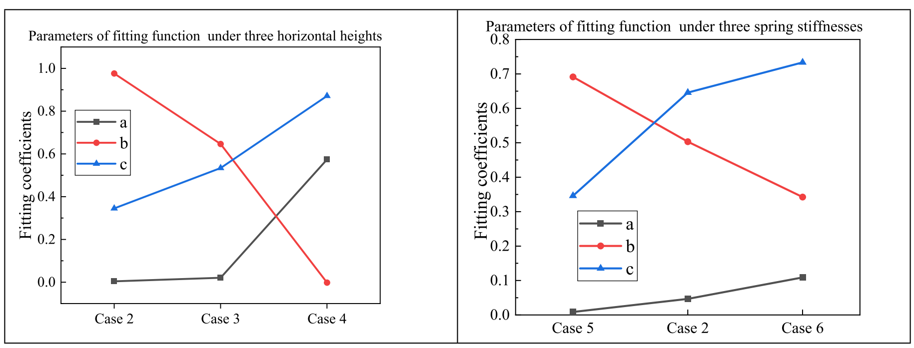

Fig. 10. Parameter values of fitting function for MGV: (a) three horizontal lengths; (b) three spring stiffnesses.

0.008,0.013 and 0.018, respectively. The elliptical shapes' areas of vertical phase planes for the Case 6 and Case 5 are the largest and the smallest, respectively.

Fig. 8 compares the control effects of the spring stiffnesses of the MGV on torsional VIV displacement responses of the bridge, and the MGV has significant control effect on the torsional VIV under three wind attack angles. It is noted that a small torsional VIV responses of about 0.04° and 0.03° for the Case 6 under the α=+3° and α = -3°, respectively. As shown in Fig. 8(d), the maximum and the minimum suppression ratios of torsional displacement are the Case 6 with 91.82 % under the α = 0° and the 77.29 % under the α = -3°, respectively. As shown in Fig. 8(e-f), the peak torsional amplitudes of the irregular sinusoidal motions for Case 2, Case 5, and Case 6 are about 0.035°,0.023° and 0.015°, respectively. The elliptical shapes' area of torsional phase planes for Case 6 is slight larger than those of Case 2 and Case 5. With the increase of the spring stiffness, the MGV has better control effect on torsional VIV responses of the closed-box girder bridge.

3.4. Movement characteristics of the moveable guide vane

As a damped least-squares method, the Levenberg–Marquardt algorithm is often used to solve non-linear least squares problems. The Levenberg-Marquardt and Universal Global Optimization (LM-UGO) method has great ability at finding the global optimal solution with fast speed and high accuracy [30], and is applied for the optimization process in establishing mapping relationship among the wind speed, the vertical displacement of the closed-box girder and the MGV, as well as their nonlinear exponential function are fitted by the Eq. (5).

$$Z = a^*X^{b*}Y^{c} \tag{5}$$

where Z is the vertical displacement of the closed-box girder, X is the wind speed, Y is the vertical displacement of the MGV, a,b,c are the fitting cofficients.

There are an obvious time lag of the vertical VIV responses for the MGV in Case 3 compared with the vertical VIV responses for the closed-box girder under the wind speeds increase from 7 m/s to 15 m/s, and the peak vertical displacements of the MGV in Case 3 are much larger than those of the closed-box girder. As shown in the Fig. 4 and Fig. 9, there are two movable guide vanes simultaneously installed at two sides of the closed-box girder in this study, and the movement of two moveable guide vanes could result in energy dissipation of the section model structure in the lock-in wind speed of VIV. Although the vertical and torsional VIV displacement of the bridge could be suppressed by the combined effect of two guide vanes' synchronous and asynchronous movement, respectivly, both the vertical and torsional VIV control mechanisms are the energy consumption through two guide vanes' large movement instead of only the main girder's movement. Under other five cases, the vertical displacement responses of the MGV gradually increase with the increase of wind speed. As a whole, the vertical displacement responses of the MGV are slight larger than those of the closed-box girder with the synchronous non-VIV motion. In the unlock-in wind speed region, the vertical displacements of the movable guide vanes are not sensitivy to the change of the horizontal lengths or the spring stiffness. Fig. 9 also shows that the nonlinear exponential fitting function can perfectly satisfy the demands since all these R-Squares (R^2) of these different cases are larger than 0.92, which indicates there are stong coupled effect among the motion of the closed-box girder and the MGV. It is intereating that the shape of the three-dimensional fitting function for the Case 3 is close to linear surface, while the shapes of the fitting functions for other five cases are the curved surface with small top and large bottom. In addition, the R^2 of the mapping relationship in Case 3 are as high as 0.095, but the R^2 of other five cases are lower than 0.095. In summary, the vertical displacement responses of the movable guide vanes are close to those of the main girder, while the movable guide vane could play an important role in the VIV suppression by the double itself motion amplitude.

Furthermore, three fitting cofficients in Eq. (5) are compared with three different horizontal lengths and the spring stiffness. Fig. 10(a) shows that the values of a and c gradually become larger with the increase of the horizontal length, which is opposite to those of the b values. Since the wind speed of × is larger than 1, the weight of the motion of the MGV on affecting the motion of the closed-box girder decreases with the increase of the horizontal length. Fig. 10(b) also shows that the values of a and c gradually become larger, while the b values gradually decrease. The resulting indicates that the effect weight of the MGV with larger stiffness on the motion of the closed-box girder become smaller.

4. Conclusion

A series of VIV tests of a closed-box girder with various moveable guide vanes are conducted to study the control effectiveness of the moveable guide vane on the VIV performance of a suspension bridge. The major findings in this study are as following:

- The moveable guide vane could significantly reduce the vertical and torsional displacement responses of the closed-box girder bridge, and the suppression ratio of torsional VIV displacement could be reach more than 77 %.
- The MGV with a relative smaller horizontal lengths or a relative larger spring stiffness has a better VIV control effect of the closed-box girder bridge, and the control effect could be unbenefit under the wind attack angles of -3°.
- The maximum vertical displcement of the moveable guide vane for the Case 3 is much larger than that of the closed-box girder in the VIV lock-in region, while the vertical displcement of the moveable guide vane for other five cases are close to those of the closed-box girder.
- The nonlinear exponential function could effectively reflect the relationship among the wind speed, the vertical displacement of MGV and the closed-box girder. With the increase of horizontal length and spring stiffness, the effect weight of the MGV on the motion of the closed-box girder become smaller.

Although the moveable guide vane with a relatively short length (*i.e.,* 20 mm) and high spring stiffness (*i.e.,* 24 g/mm) could enhance VIV performance of the closed-box girder suspension bridge, the influence of the optimal movable guide vanes on the flutter and buffeting performance could be further studied in future.

CRediT authorship contribution statement

Rui Zhou: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Yiyao Zhang:** Software, Formal analysis. **Yaojun Ge:** Writing – review & editing, Supervision. **Yongxin Yang:** Project administration, Funding acquisition, Visualization. **Xiaodong Gao:** Investigation, Data curation. **Peng Lu:** Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the support for the research work jointly provided by the Guangdong Natural Science Foundation (No.2023A1515030148), the National Natural Science Foundation of China (Nos. 52278311, 52178503, U2005216, and 51908374), and the Shenzhen Science and Technology Program under grant (Nos. JCYJ20220531101609020 and KQTD20200820113004005, GJHZ20220913143006012), Foundation of State Key Laboratory for Disaster Reduction in Civil Engineering, Tongji University (No. SLDRCE19-B-10), and the National Key Laboratory of Green and Long-Life Road Engineering in Extreme Environment.

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